

Team name:	#REF!
------------	-------

Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Korean Won KRW	1,888,590	\$1,259.06

#	Component	Part name	Author	Source	Quantity	Unit cost (Local Currency)	Unit cost (US Dollars)	Total Cost (Local Currency)	Total Cost (USA Dollars)
1	Main controller Core S3 lite	M5Stack CoreS3 Lite (ESP32-S3, 9-axis IMU BMI270+BMM150)	M5Stack	shop.m5stack.com	1	82,000	\$54.67	82,000	\$54.67
2	BaseX	M5Stack BaseX (SKU K037, STM32 motor base, I2C 0x22)	M5Stack	shop.m5stack.com	1	42,550	\$28.37	42,550	\$28.37
3	Drive motor	LEGO MINDSTORMS EV3 Medium Servo Motor (45503)	LEGO	LEGO Education	4	160,000	\$106.67	640,000	\$426.67
4	Reduction gear (driver)	LEGO Spur Gear 12T, module 0.5	LEGO	BrickLink	4	2,000	\$1.33	8,000	\$5.33
5	Reduction gear (driven)	LEGO Spur Gear 28T, module 0.5	LEGO	BrickLink	4	2,000	\$1.33	8,000	\$5.33
6	Tire	LEGO Tire 62.4mm D. x 20mm (part 32019)	LEGO	BrickLink	4	42,000	\$28.00	168,000	\$112.00
7	Wheel / rim	LEGO Wheel 62.4 x 20 (part 32020 / 86652)	LEGO	BrickLink	4	12,800	\$8.53	51,200	\$34.13
8	Distance sensor (ToF)	VL53L0/1XV2		AliExpress	4	17,760	\$11.84	71,040	\$47.36
9	Color sensor	M5Stack Unit Color (TCS3472 RGB)	M5Stack	shop.m5stack.com	2	35,600	\$23.73	71,200	\$47.47
10	I2C multiplexer	M5Stack Unit PaHub (PCA9548A, 1-to-6)	M5Stack	shop.m5stack.com	1	25,000	\$16.67	25,000	\$16.67
11	Camera	M5Stack UnitV2 M12 (edge camera)	M5Stack	shop.m5stack.com	2	301,000	\$200.67	602,000	\$401.33
12	Dropper servo	AGFRC C017CLS 1.7g micro servo	AGFRC	agfrc.com	2	17,800	\$11.87	35,600	\$23.73
13	Chassis & structure	LEGO Technic beams/axles/connectors (custom frame)	LEGO parts / Top Class (assembly)	BrickLink	1	5,000	\$3.33	5,000	\$3.33
14	Differential & suspension	LEGO Technic differential gearset + parts	LEGO parts / Top Class (assembly)	BrickLink	1	8,000	\$5.33	8,000	\$5.33
15	3D-printed parts (PLA)	Custom mounts/brackets/dropper housing, PLA filament	Top Class (team, designed in Fusion 360)	3D printer / self-made	1	21,000	\$14.00	21,000	\$14.00
16	Motor power battery	External DC battery pack 9-12V (motor power)	(verify)	local store	1	26,800	\$17.87	26,800	\$17.87
17	Camera power supply	5V BEC / battery for UnitV2 (>=2A)	(verify)	local store	1	15,200	\$10.13	15,200	\$10.13
18	Cables & connectors	HY2.0 Grove cables / RJ12 / wiring (lot)	M5Stack / local	shop.m5stack.com	1	8,000	\$5.33	8,000	\$5.33

Team name:	#REF!
------------	-------

Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Korean Won	R\$0.00	\$0.00

#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Local Currency)	Cost (USA Dollars)
1	MicroPython (M5Stack UIFlow2)	Software's Tool	Robot firmware runtime on Core S3	github.com/m5stack/uiflow-micropython	M5Stack / Damien George	FREE	FREE
2	Visual Studio Code	Software's Tool	Code editor / upload	code.visualstudio.com	Microsoft	FREE	FREE
3	Fusion 360 (Personal Use)	Software's Tool	3D CAD modeling of custom printed parts	autodesk.com/fusion-360	Autodesk	FREE	FREE
4	mpremote	Software's Tool	Board file transfer & run	github.com/micropython/micropython	MicroPython (George Robotics)	FREE	FREE
5	uasyncio	Library	Async cooperative multitasking	MicroPython builtin	MicroPython	FREE	FREE
6	BMI270 + BMM150 driver	Library	9-axis IMU driver	m5stack micropython drivers	M5Stack	FREE	FREE
7	VL53L0X driver	Library	ToF distance sensor driver	github.com/pololu/vl53l0x-arduino	Pololu	FREE	FREE
8	TCS3472 driver	Library	Color sensor driver	github.com/adafruit/Adafruit_TCS34725	Adafruit	FREE	FREE
9	PCA9548A (PaHub) driver	Library	I2C multiplexer channel control	m5stack micropython	M5Stack	FREE	FREE
10	DriveBase / Motor library	Library	Pybricks-style BaseX motor control (self-developed)	self-made	Top Class (team)	FREE	FREE
11	IMU fusion (fusion / imu_opt)	Library	Gyro drift correction & heading (self-developed)	self-made	Top Class (team)	FREE	FREE
12	Maze explorer (maze_map / explorer)	Library	Maze mapping & path search (self-developed)	self-made	Top Class (team)	FREE	FREE
13	UnitV2 vision (letters / color ring)	AI model	Victim & cognitive-target recognition (self-developed)	self-made / V-Training	Top Class (team)	FREE	FREE
14	UnitV2 onboard firmware	Software's Tool	Camera edge-AI firmware	docs.m5stack.com/en/unit/unitv2	M5Stack / SigmaStar	FREE	FREE
15	Python 3 (desktop)	Software's Tool	Desktop simulation & verification	python.org	Python Software Foundation	FREE	FREE
16	NumPy	Library	Numerical computation in simulation	numpy.org	NumPy developers	FREE	FREE